

File Inventory — EMS Readiness Optimization Project

Generated: 2026-03-15 | Version: 1.3.0 | Total tracked files in remote: **230+** | Total local files: **240+**

Phase Deliverable Checklist

Phase	Description	Status
Phase 1	Data Processing & EDA	✓ Complete
Phase 2	Demand & Service Modeling (NHPP)	✓ Complete
Phase 3	Optimization Models (P0, P1, P2)	✓ Complete
Phase 4	DES Simulation with V&V (39 tests)	✓ Complete
Phase 5	Experimental Design & Production Runs (1,440 sims)	✓ Complete
Phase 6	Comprehensive Statistical Analysis	✓ Complete
Phase 7	Final Report & Documentation	✓ Complete
Phase 8	Alternative Analyses & Extensions	✓ Complete
Phase 9	Capacity & Baseline Improvements	✓ Complete

Directory Structure & File Inventory

configs/ — Configuration Files

File	Size	Description
demand.yaml	856 B	NHPP demand model parameters
optimization.yaml	960 B	MIP solver and unit-count settings (capacity=2 default)
service.yaml	1.5 KB	Travel time and service time config
simulation.yaml	1.1 KB	DES engine and batch runner settings
cbd_scenario.yaml	~500 B	CBD robustness experiment config

data/raw/ — Raw Input Data

File	Size	Tracked	Description
FDNY_Firehouse_Listing_20260223.csv	34 KB	✓	219 FDNY firehouses
FDNY_Firehouse_Listing_Data_Dictionary.xlsx	66 KB	✓	Firehouse field definitions
Motor_Vehicle_Collisions_-_Crashes_20260223.csv	536 MB	✗	2.24M crash records (too large)
Motor_Vehicle_Collisions_Data_Dictionary.xlsx	96 KB	✓	Crash field definitions
Precincts_20260223.csv	3.6 MB	✗	78 precinct boundaries (large)
Precincts_Data_Dictionary.xlsx	62 KB	✓	Precinct field definitions
manhattan_boundary.geojson	247 KB	✓	Manhattan polygon boundary
cbd_boundary.geojson	101 KB	✓	CBD/Congestion Relief Zone boundary
nyc_borough_boundaries.geojson	3.1 MB	✓	All 5 NYC borough boundaries

data/processed/ — Processed Data

File	Size	Tracked	Description
crashes_manhattan.csv	99 MB	✗	Filtered Manhattan crashes (large)
crashes_manhattan.parquet	15 MB	✗	Parquet version of above
de-mand_lambda_hourly.csv	1.9 KB	✓	Hourly λ factors (24 hours)
de-mand_lambda_dow.csv	653 B	✓	Day-of-week λ factors
de-mand_lambda_precinct.csv	1.7 KB	✓	Precinct-level λ rates
de-mand_model_summary.json	696 B	✓	Demand model summary stats
distance_matrix_firehouse_precinct.csv	27 KB	✓	48×30 Haversine distance matrix
distance_matrix_firehouse_precinct_manhattan.csv	27 KB	✓	48×30 Manhattan distance matrix
fire-houses_clean.csv	32 KB	✓	All cleaned firehouses
fire-houses_manhattan.csv	7.1 KB	✓	48 Manhattan firehouses
pre-cincts_manhattan.geojson	640 KB	✓	30 Manhattan precincts

src/ems_readiness/ — Source Code (8,500+ lines total)

Module	Files	Description
<code>__init__.py</code>	1	Package metadata (v0.6.0)
<code>demand/</code>	<code>__init__.py</code> , <code>arrival_generator.py</code>	NHPP arrival generation (thinning algorithm)
<code>service/</code>	<code>__init__.py</code> , <code>travel_time.py</code> , <code>service_time.py</code>	Travel proxy & LogNormal service time
<code>optimization/</code>	<code>__init__.py</code> , <code>models.py</code> , <code>policies.py</code> , <code>allocator.py</code>	MIP formulations (incl. CBD-focused, capacity-aware) & baseline policies (incl. spatially-stratified)
<code>simulation/</code>	<code>__init__.py</code> , <code>engine.py</code> , <code>entities.py</code> , <code>resources.py</code> , <code>dispatcher.py</code> , <code>metrics.py</code> , <code>runner.py</code>	SimPy DES engine with batch runner
<code>utils/</code>	<code>__init__.py</code> , <code>distance.py</code>	Haversine & Manhattan distance calculations

scripts/ — Automation Scripts (24 scripts)

File	Phase	Description
data_audit.py	2	Data quality audit pipeline
audit_step1_boundaries.py	2	Boundary file validation
audit_step2_firehouses.py	2	Firehouse data validation
audit_step3_precincts.py	2	Precinct data validation
audit_step4_crashes.py	2	Crash data validation
demand_modeling.py	2	NHPP demand model fitting
run_optimization_comparison.py	3	Multi-model optimization comparison
run_verification.py	5	Simulation verification tests
run_validation_pilots.py	5	Validation pilot experiments
run_production_experiments.py	6	Full 1,440-run production experiments
analyze_production_results.py	6	Production result analysis
generate_publication_figures.py	6	Publication-quality figure generation
generate_summary_dashboard.py	6	Project summary dashboard
run_cbd_experiment.py	6	CBD robustness experiment (330 runs)
analyze_queue_metrics.py	6	Queue metrics analysis
analyze_seasonal_patterns.py	6	Seasonal variation analysis
generate_manhattan_distance_matrix.py	8	Manhattan (taxicab) distance matrix generation
run_distance_comparison_experiment.py	8	Haversine vs. Manhattan metric comparison
run_cbd_focused_optimization.py	8	CBD-focused vs. Manhattan-wide optimization

File	Phase	Description
capacity_sensitivity_analysis.py	9	Initial cap=2 vs cap=5 comparison
capacity_sensitivity_full_spectrum.py	9	Full-spectrum cap 1-5 sensitivity (450 runs)
p0_spatial_analysis.py	9	P0 spatial stratification analysis and visualization
run_production_v2.py	9	Production V2 experiment runner (810 runs)
generate_all_heatmaps.py	9	Generates 108 staging location heat maps (9K × 3 policies × 4 capacities)

tests/ — Test Suite (39 tests)

File	Tests	Description
conftest.py	—	Shared fixtures
test_simulation_core.py	15	Core DES engine tests
test_dispatch_logic.py	10	Dispatcher & resource tests
test_extreme_cases.py	8	Edge case & stress tests
test_reproducibility.py	6	Seeded reproducibility tests

notebooks/ — Analysis Notebooks (8 notebooks)

File	Description
02_eda_spatiotemporal.ipynb	Exploratory data analysis
03_input_modeling.ipynb	NHPP demand model fitting
04_service_travel_proxy.ipynb	Service & travel time validation
05_optimization.ipynb	Optimization model comparison
06_simulation_debug.ipynb	Simulation debugging & V&V
07_production_results.ipynb	Production experiment results
08_statistical_analysis.ipynb	Comprehensive statistical analysis
09_cbd_analysis.ipynb	CBD robustness analysis

results/figures/ — Visualizations (66+ PNG files)

Category	Files	Description
EDA	10	Crash heatmap, temporal trends, firehouses map, etc.
Optimization	4	Inputs, sensitivity, allocation comparison
Simulation V&V	4	Verification timeline, validation comparisons
Experiments	4	Exp1-Exp4 result plots
Publication	5	pub_fig1-pub_fig5 (high quality)
Model	6	Distance heatmap, travel time, service time, NHPP
CBD	3	CBD robustness figures
Queue	4	Queue analysis figures
Seasonal	3	Seasonal analysis figures
Distance Comparison	4	Haversine vs Manhattan figures
CBD-Focused Comparison	3	CBD-focused vs Manhattan-wide figures
Capacity Sensitivity	10+	Cap 1-5 sensitivity figures
Production V2	10+	V2 result figures

results/tables/ — Statistical Tables (55+ files)

File	Description
descriptive_statistics.csv	Summary stats for all experiments
anova_results.csv	ANOVA test results
posthoc_comparisons.csv	Pairwise comparisons (Tukey HSD)
confidence_intervals.csv	95% confidence intervals
effect_sizes.csv	Cohen's d effect sizes
sensitivity_summary.csv	Sensitivity analysis summary
table1-table4_*.csv/.tex	Publication-ready tables (CSV + LaTeX)
exp1-exp4_*.csv	Per-experiment pivot tables
optimization_comparison.csv	Policy comparison metrics
cbd_*.csv	CBD robustness tables
queue_*.csv	Queue analysis tables
seasonal_*.csv	Seasonal analysis tables

results/simulation/ — Simulation Outputs

Directory	Files	Description
production/	6	4 experiment CSVs + summary + log
validation_pilot/	5	3 pilot JSONs + 2 comparison tables
verification/	4	4 verification test JSONs

results/optimization/ — Optimization Results

File	Description
allocations_K20–K48.csv	Unit allocations for K=20,30,40,48
policy_comparison.csv	5-policy comparison results
sensitivity_analysis.csv	K-sensitivity analysis
findings_summary.json	Machine-readable summary
PHASE3_SUMMARY.md/.pdf	Phase 3 narrative summary

results/distance_comparison/ — Distance Metric Comparison (Phase 8)

File	Description
comparison_table.csv	Haversine vs. Manhattan simulation metrics
allocation_comparison.csv	Side-by-side firehouse allocations
distance_matrices_heatmap.png	Dual-panel heatmap
distance_scatter.png	Manhattan vs. Haversine scatter
distance_comparison_bar.png	RT comparison bar chart
distance_comparison_boxplot.png	RT distribution boxplot
experiment_log.txt	Experiment execution log

results/cbd_focused_comparison/ — CBD-Focused Optimization (Phase 8)

File	Description
comparison_table.csv	CBD-focused vs. Manhattan-wide metrics
allocations.csv	Side-by-side allocations
cbd_focused_comparison.png	RT comparison bar chart
allocation_comparison.png	Allocation distribution comparison
equity_tradeoff.png	CBD vs. non-CBD equity trade-off
experiment_log.txt	Experiment execution log

results/capacity_comparison/ — Capacity Sensitivity Analysis (Phase 9)

File	Description
cap_*_allocations_*.csv	Allocation vectors per capacity $\times$ K
capacity_sensitivity_summary.csv	Summary table: cap 1-5 $\times$ K 10-48
capacity_*_figures.png	Capacity sensitivity visualization figures
experiment_log.txt	Experiment execution log

results/production_v2/ — Production V2 Results (Phase 9)

File	Description
v2_allocations_*.csv	P0-spatial, P1, P2 allocations per K
v2_simulation_results.csv	Full simulation results (810 runs)
v2_summary_statistics.csv	Summary statistics by policy $\times$ K
comparison_with_v1.csv	V1 vs V2 comparison table
v2_*.png	Production V2 result figures

results/heatmaps/ — Staging Location Heat Map Collection (108 maps)

A systematic collection of **108 heat maps** visualizing ambulance staging locations on Manhattan geography for every combination of fleet size, allocation policy, and per-firehouse capacity limit. These maps are generated by `scripts/generate_all_heatmaps.py`.

Parameter space:

Parameter	Values	Count
Fleet size (K)	5, 10, 15, 20, 25, 30, 35, 40, 45	9
Policy	P0-spatial, P1, P2	3
Capacity	1, 2, 3, 5	4
Total	9 $\times$ 3 $\times$ 4	108

Naming convention: heatmap_K{k}_policy{policy}_cap{capacity}.png

Example filename	Description
heatmap_K20_policyP2_cap2.png	Optimized allocation (P2), 20 units, capacity 2
heatmap_K10_policyP0_spatial_cap1.png	Spatial baseline (P0), 10 units, capacity 1
heatmap_K40_policyP1_cap5.png	Demand-proportional (P1), 40 units, capacity 5

Each map shows active staging locations (sized/colored by unit count) overlaid on the Manhattan borough boundary with precinct outlines and CBD boundary for reference. The companion `allocations/` subdirectory contains the underlying allocation vectors as CSV files.

These maps support scenario exploration across the full experimental space and feed into the interactive dashboard.

docs/ — Documentation (32+ files)

File	Description
project_charter.md	Project scope and objectives
source_manifest.md/.pdf	Data source inventory
assumptions_log.md	Documented assumptions (14 items)
decisions_log.md/.pdf	Key decision records (DEC-001 through DEC-011)
blocker_log.md	Issue tracking
demand_model_spec.md/.pdf	NHPP demand model specification
service_model_spec.md/.pdf	Travel & service model spec
optimization_formulation.md/.pdf	MIP formulation documentation
optimization_results.md/.pdf	Optimization results narrative
conceptual_model.md/.pdf	DES conceptual model
experimental_design.md/.pdf	Factorial experimental design
verification_log.md/.pdf	V&V test results
output_analysis.md/.pdf	Statistical analysis report
executive_summary.md/.pdf	Executive summary
technical_report.md	Full final report (v2.1.0)
executive_presentation.md	Slide deck for stakeholders
implementation_roadmap.md	Deployment plan
project_archive.md	Project archive & timeline
code_documentation.md	Code architecture guide (v1.3.0)
file_inventory.md	This file
cbd_robustness_analysis.md	CBD robustness DES experiment
queue_analysis.md	Queueing performance analysis
gap_closure_report.md	Gap closure verification
distance_metric_comparison.md/.pdf	

File	Description
	Haversine vs. Manhattan comparison (Phase 8)
<code>cbd_focused_optimization_analysis.md/.pdf</code>	CBD-focused optimization report (Phase 8)
<code>alternative_analyses_summary.md/.pdf</code>	Alternative analyses summary (Phase 8)
<code>research_questions_assessment.md</code>	Research questions assessment
<code>project_workflow_wbs.md/.pdf</code>	Work Breakdown Structure (v1.3.0)
<code>firehouse_capacity_analysis.md</code>	Firehouse capacity methodology (Phase 9)
<code>capacity_sensitivity_analysis.md</code>	Full-spectrum capacity sensitivity report (Phase 9)
<code>final_summary.md</code>	Final project summary (v1.3.0)

Project Statistics

Metric	Value
Total Python LOC	8,500+
Total source modules	14
Total scripts	24
Total test cases	39
Total notebooks	8
Total figures	72+ (analysis) + 108 (staging heat maps)
Total CSV result files	55+
Total documentation files	32+
Total simulation runs	2,700+ (production V1 + V2 + CBD + capacity + alternatives)
Tracked files in Git	230+
Project size (excl. raw data)	~160 MB

